HAPPIEST HEALTH DIGITAL TWIN CHALLENGE 2026

Research prototype · not for clinical use

A glycaemic digital twin that warns of highs and lows two hours ahead

Type 2 diabetes · health record + wearables · explained, personalised, tested on real patients



Sync & Predict · BITSoM

Highs and lows happen between clinic visits

101 M

people with diabetes in India, mostly Type 2 (ICMR-INDIAB, Lancet Diabetes & Endocrinology 2023)

3 months

between HbA1c tests, while dangerous highs and lows happen within hours

At night

lows from sulfonylureas or insulin strike while the patient sleeps and cannot react

Health record
diagnoses, labs, medicines

never combined

Wearable sensor
glucose every 5 minutes

So nobody turns them into an early warning a doctor can act on.

A digital twin of each patient

01

Predict

Risk of glucose above 180 or below 70 mg/dL in the next 2 hours, every 30 minutes

02

Alert

One alert per risk episode, tuned so clinicians are not flooded with alarms

03

Explain

The top reasons behind every risk: falling glucose, a sulfonylurea dose, the time of day

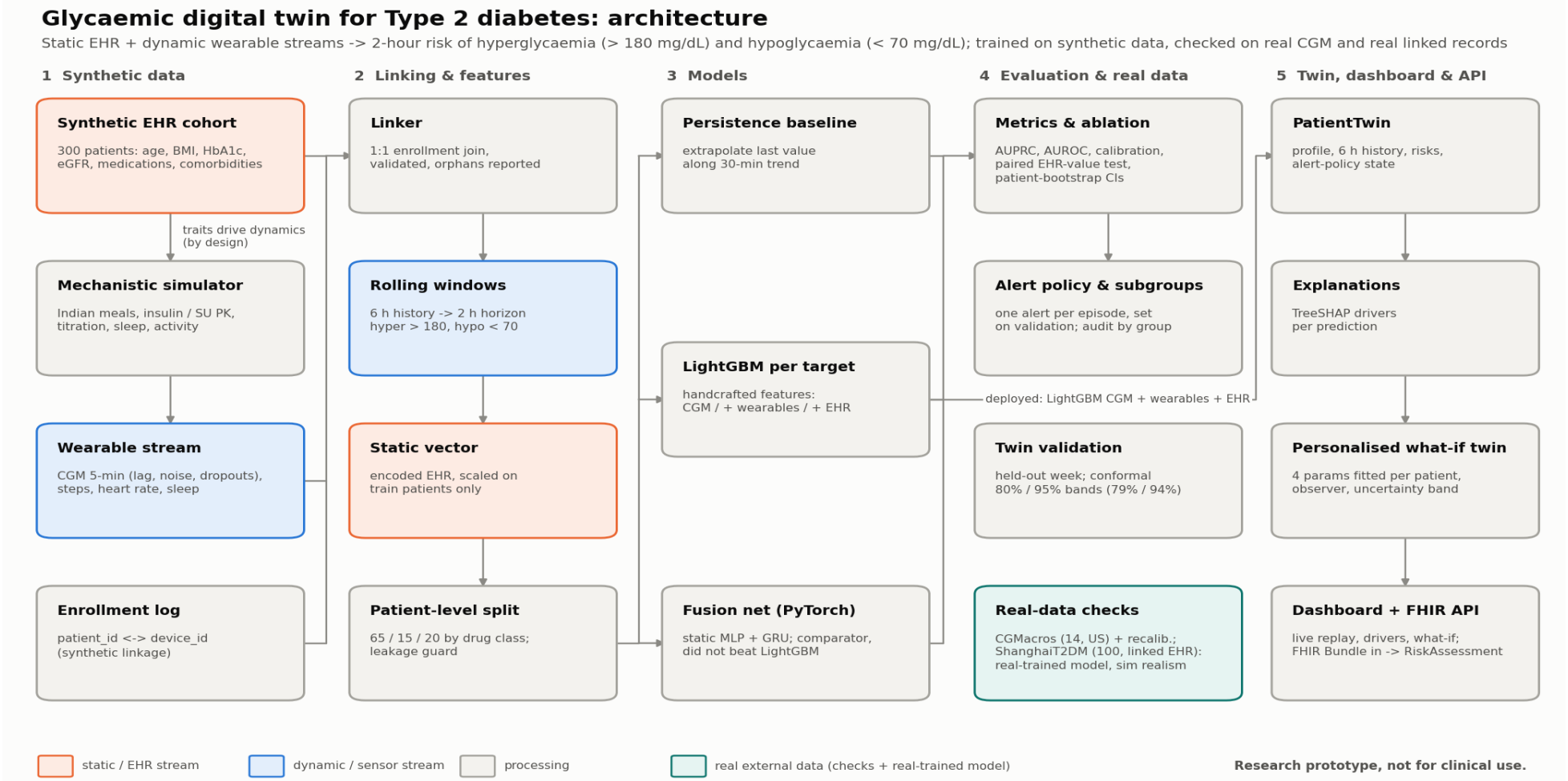
04

Simulate

A twin fitted to the patient's own data answers 'what if?' with an honest uncertainty band

Inputs: the last 6 hours of CGM, steps, heart rate and sleep, plus the health record. Output: an explained 2-hour early warning.

Five layers, from data to dashboard and API



A physiology simulator, honestly labelled

What we generate

300 patients × 14 days
5-minute CGM with sensor lag, noise and dropouts

A full health record
age, BMI, HbA1c, eGFR, lipids, comorbidities, drug doses

Indian-context physiology
rice, roti, idli, poha; drug pharmacokinetics; sleep, activity, dawn effect

Split by patient
194 train / 46 validation / 60 test, stratified by drug class

Synthetic by design, and labelled

No public dataset links hospital records and wearables for the same Indian patients, so we simulate both.

The simulator makes EHR traits drive glucose, so any EHR gain on this data is partly built in. **That is why we went on to test the EHR on real linked records.**

Built so the numbers can be trusted

01

Patient-level split

No patient in both training and test; the code raises an error if one appears

02

Onset windows

Predictions are scored before the event starts, never once glucose is already low

03

Paired bootstrap

Every comparison has a 95% confidence interval over patients

04

Choices on validation

Alert thresholds and policy never see the test patients

05

Reproducible

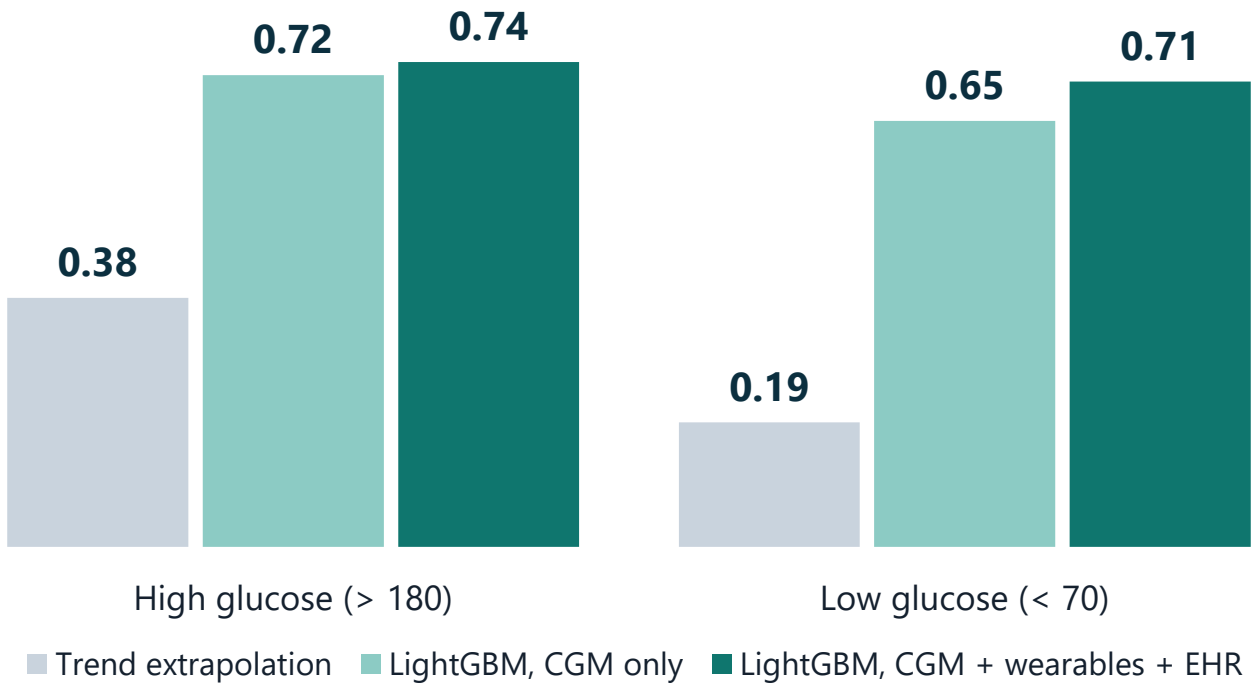
Fixed seeds, one command rebuilds everything, 140 tests on every push

Splitting by patient and scoring only onset windows remove the two most common ways glucose models look better than they are.

The model predicts both events well

Precision-recall (AUPRC) on 60 test patients

Higher is better. For rare lows, chance is about 0.02.

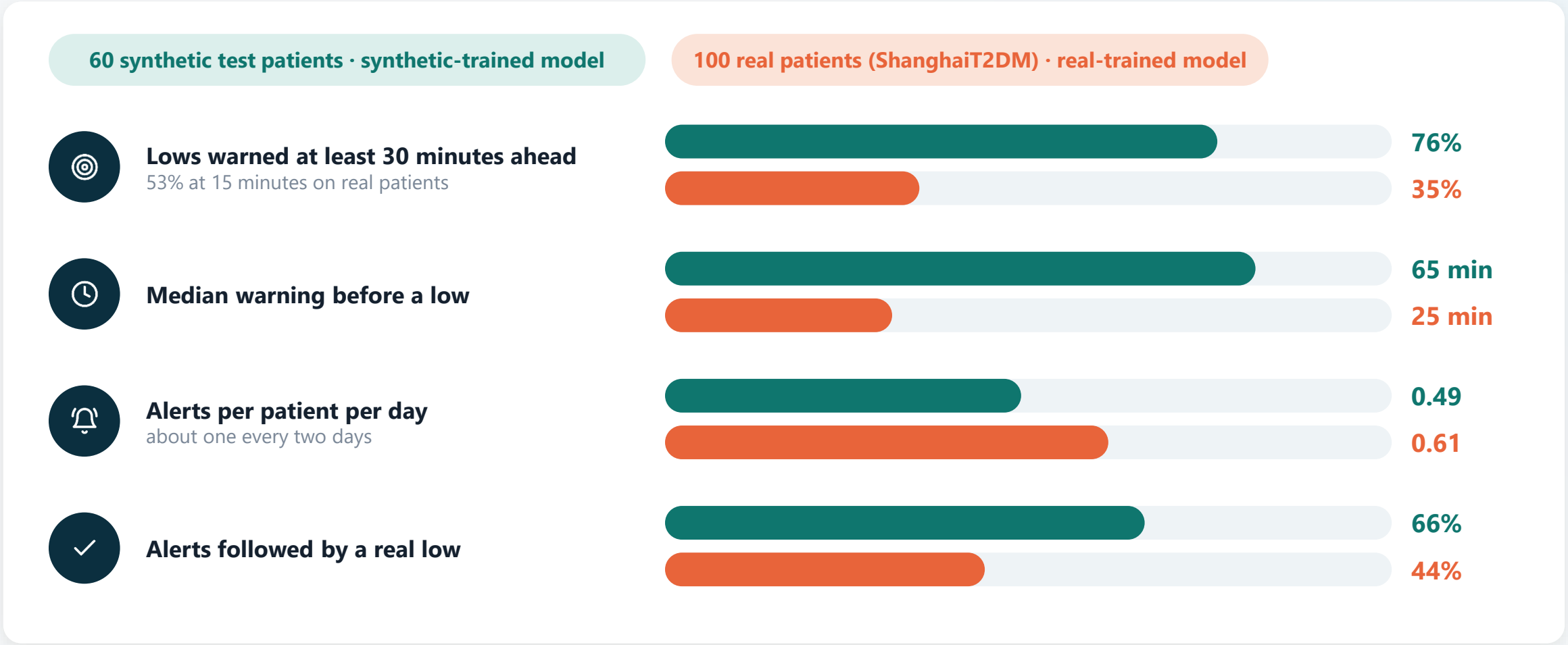


0.93
AUROC for highs

0.99
AUROC for lows

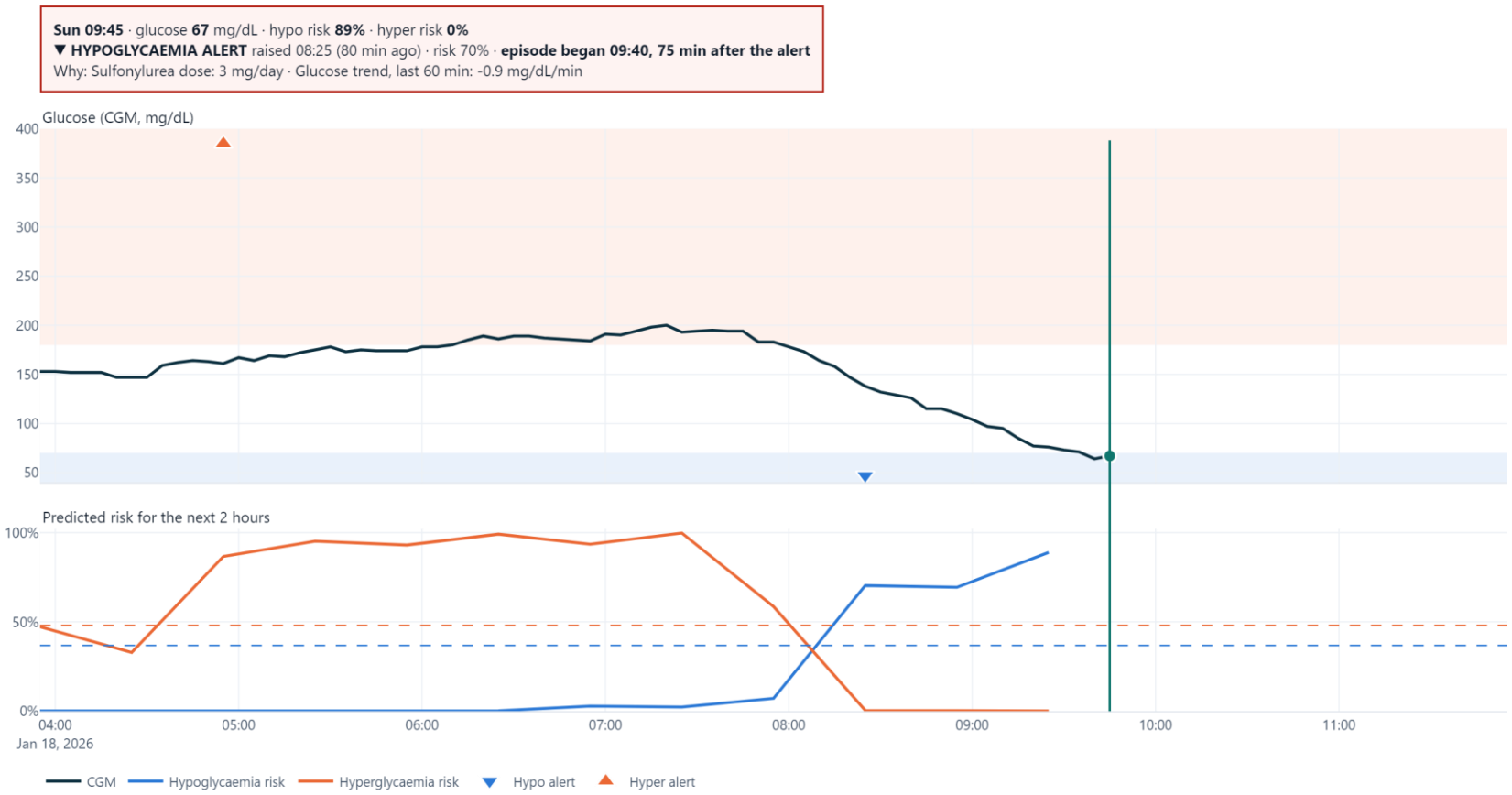
- A deep GRU fusion network did not beat LightGBM
- Steps, heart rate and sleep added nothing measurable
- Well calibrated (hypo ECE 0.004), so LightGBM is deployed
- Optimistic: simulated lows drift in slowly

Alerts a clinic can live with, tested on real patients



Real lows arrive faster than simulated ones, so the synthetic figure is an upper bound. The real-patient numbers are the honest baseline to improve.

The alert arrives 75 minutes before the low



08:25
Hypo alert at 70% risk

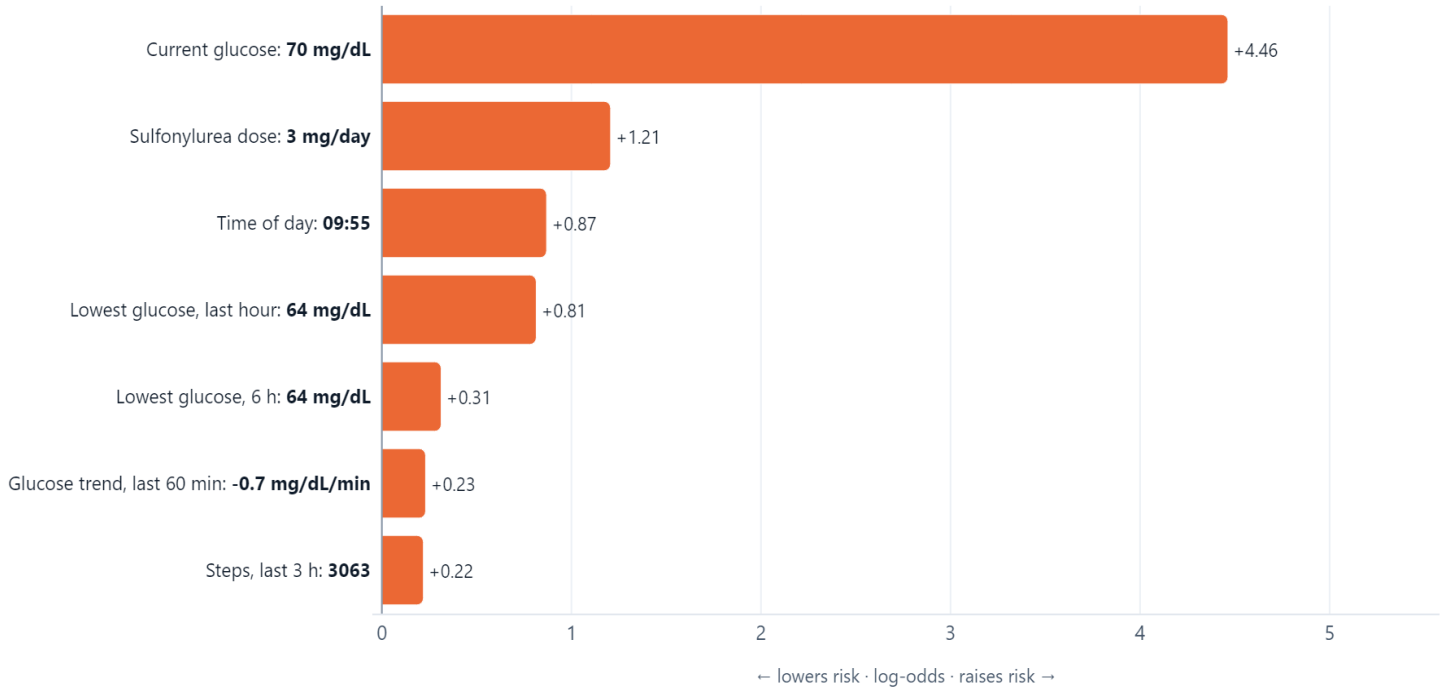
75 min warning

Why: sulfonylurea dose; glucose falling 0.9 mg/dL per minute

09:40
Glucose falls below 70

Synthetic patient P0014. The model sees only data up to the moving line.

Every alert explains itself

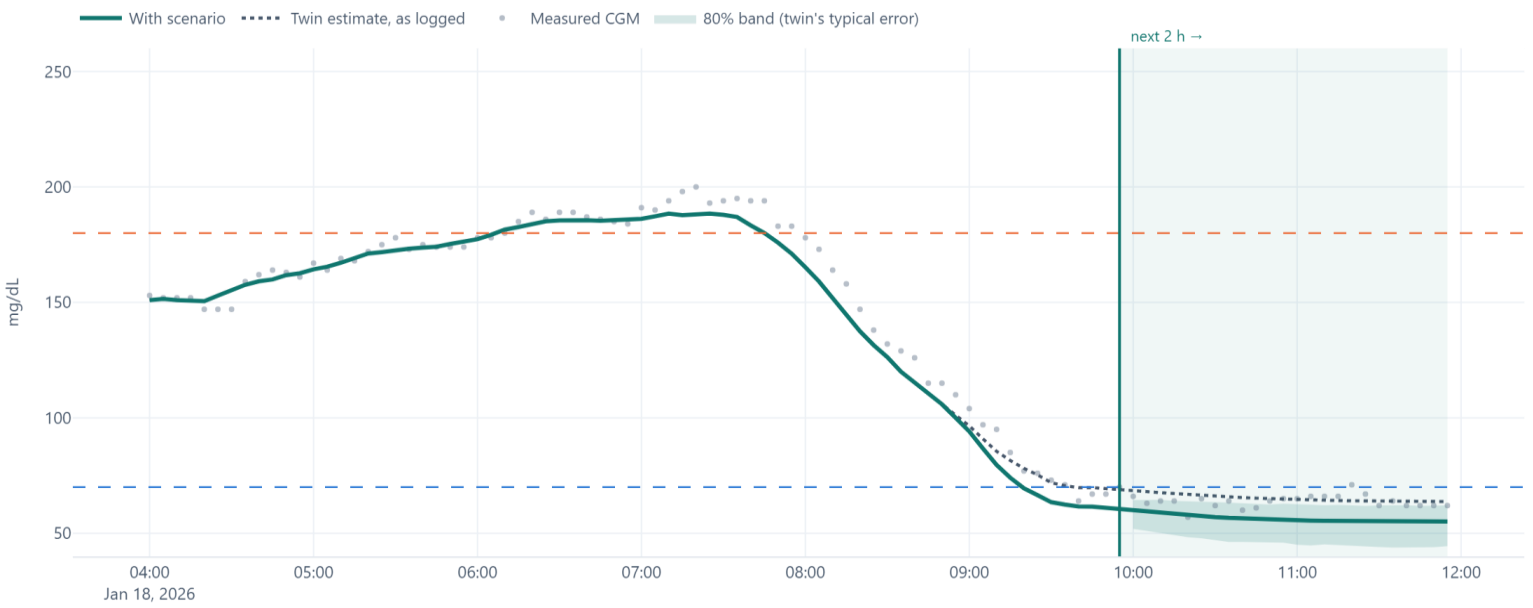


Hypoglycaemia risk 95% for synthetic patient P0014: current glucose, the sulfonylurea dose and the time of day push it up.

TreeSHAP drivers

- Exact contribution of each input, for this patient at this moment
- Grouped by data stream: CGM, time, activity and sleep, EHR
- Written as plain sentences for the clinician
- They explain the model, not the physiology, and the dashboard says so

Ask the twin 'what if?'



How it works: four physiology parameters are fitted to the patient's first week (basal glucose, carb sensitivity, clearance, drug sensitivity); a state observer tracks the CGM up to now; scenarios are re-simulated with a calibrated band. Shown: an extra 8 U insulin dose pushes the projected low from 64 to 55 mg/dL.

5.9 vs 12.1

mg/dL error on a held-out week: fitted twin vs EHR defaults

79%

of readings inside the 80% band, on patients the band never saw

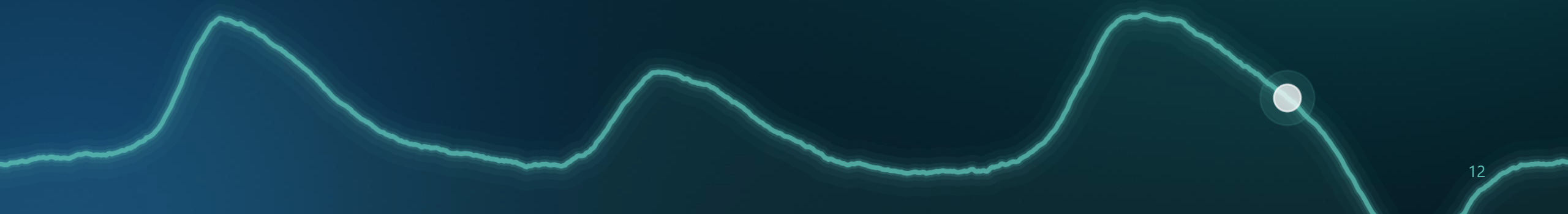
PART TWO

Then we tested it on real people

Two public datasets of real patients. Both stay outside the repository.

CGMacros · 14 US adults with T2D

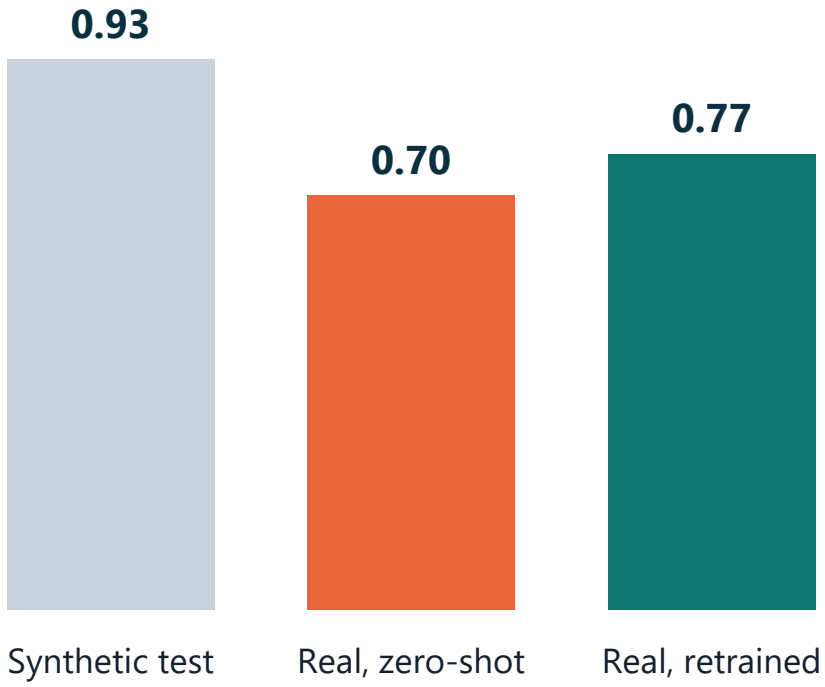
ShanghaiT2DM · 100 patients, CGM + their own records



Performance drops on real CGM, and we show it

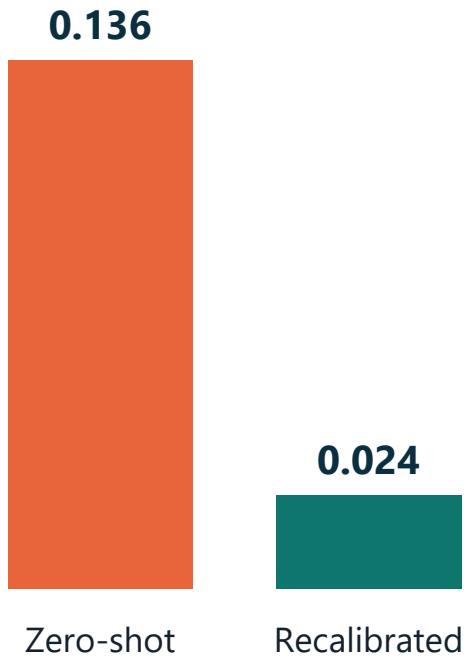
Predicting highs: AUROC

CGMacros, 14 adults with T2D. 0.5 is chance.



Calibration error

Lower is better



Honest gap

Real lows fall about twice as fast as simulated ones, and real swings are bigger

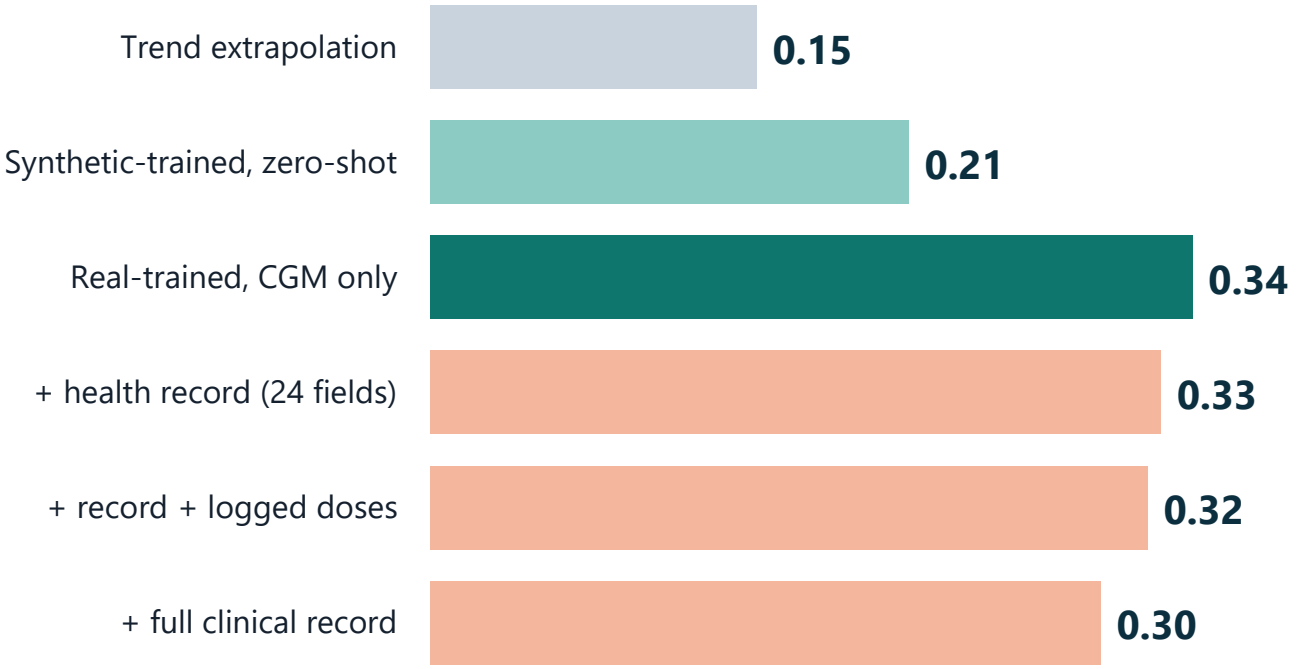
Fixable

Recalibrating on other real patients makes the probabilities trustworthy, without retraining

The CGM model predicts real lows well

Predicting real lows: AUPRC

ShanghaiT2DM, 100 patients, patient-grouped cross-validation. Chance is 0.027.



0.89

AUROC for real lows; 0.91 at night; well calibrated

+0.07

the synthetic-trained model beats trend extrapolation on real lows

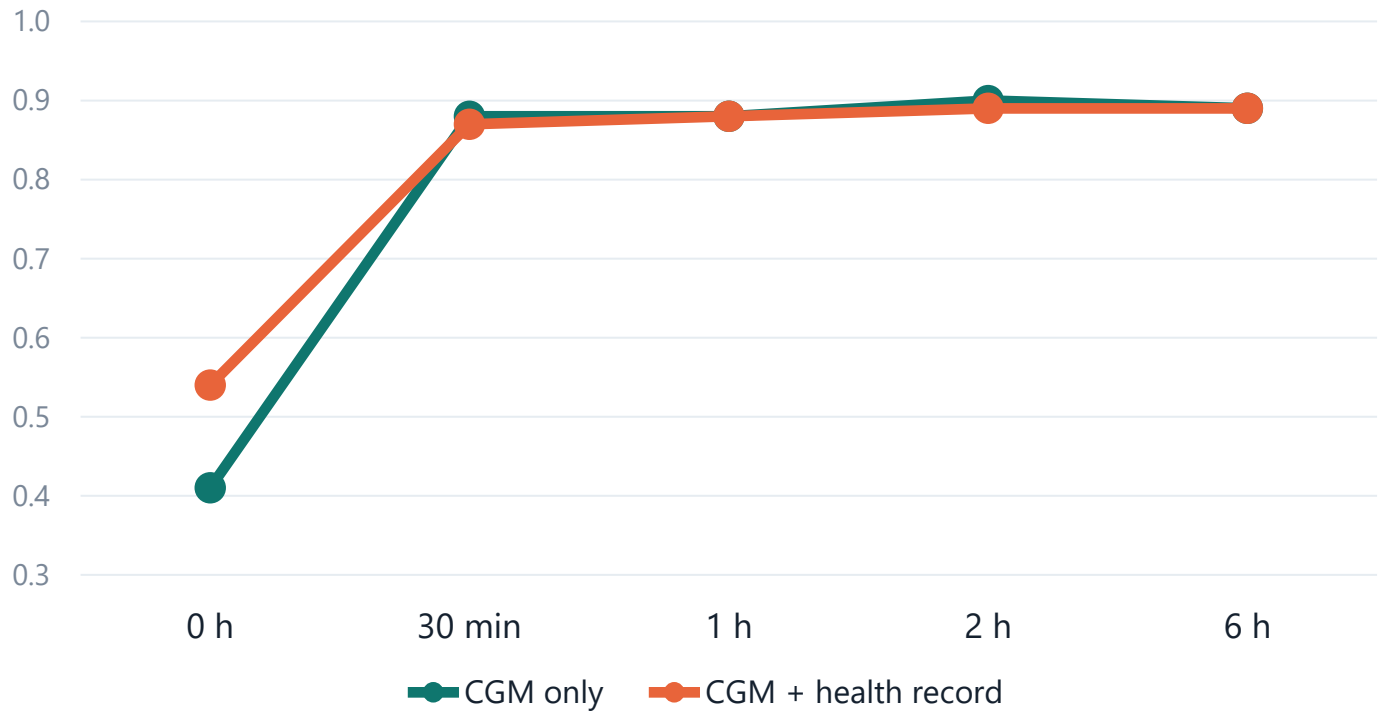
No gain

from the health record, logged doses or the full clinical record

The health record helps only before the sensor starts

Predicting real lows, by hours of CGM available: AUROC

ShanghaiT2DM. CGM alone vs CGM + 8 core record fields (low-capacity model).



No CGM yet

0.41 → 0.54

The record carries a weak but real signal: enough to decide who to watch

30 minutes of CGM

0.88

Almost as good as 6 hours, and from then on the record adds nothing

So the record is a prior, not a forecaster

What we claim, and what we no longer claim

We claim

- A CGM-based twin predicts real lows 2 hours ahead (AUROC 0.89 on 100 real patients)
- On real patients, alerts warn of 35% of lows 30 minutes ahead at 0.6 a day: an honest baseline
- Recalibration on local data fixes the probabilities
- A personalised twin halves projection error, with honest bands
- The product ships a model trained on real patients

We no longer claim

- That adding the health record improves the 2-hour forecast: true on synthetic data only
- That synthetic accuracy (0.93) is what real patients would see
- Any clinical validity: this is a research prototype

We changed our own headline claim when real data disagreed with the simulator.

Path to real use in India

STEP 1

Real-data study

Linked records and CGM from Indian T2D clinics, with ethics approval (ICMR AI guidelines)

STEP 2

Recalibrate

Fit probabilities and alert thresholds to the local population

STEP 3

Silent pilot

The model scores in the background; outcomes are compared afterwards

STEP 4

Clinician pilot

Diabetologists judge usefulness, workload and safety

- DPDP Act 2023 & Rules 2025: consent, purpose limits, security
- ABDM / ABHA consent flows to link a record to a device
- CDSCO medical device software rules (risk class to confirm)
- CGM alarms stay on; the twin never replaces them

Built to be checked, not just demoed

140
automated tests, run by CI on every push

1
command rebuilds data, models and reports

2
public real datasets checked

Live
dashboard on Streamlit Cloud

Interoperable: a FHIR R4 API takes a patient's record and CGM and returns standard RiskAssessment resources, checked to reproduce the dashboard exactly. Also: a measured comparison of the simulator with real CGM, MIT licence, model card, deployment path.

Python · LightGBM · PyTorch · SHAP · SciPy · scikit-learn · Plotly · Streamlit · FastAPI · FHIR R4

Limitations we know about

Little real training data

The default model is synthetic-trained; the real-trained one has seen only 100 patients

Not the target population

14 US and 100 Chinese patients, not Indian outpatients

Insulin users

Their lows are the hardest to predict, on synthetic and real data alike

Inputs a real twin needs

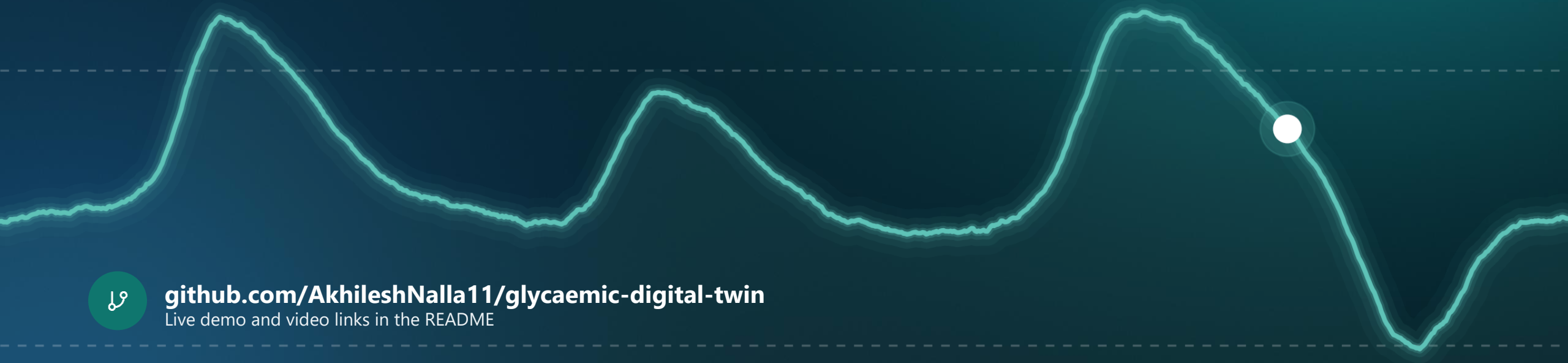
Planned meals and fresh doses are invisible to the model until glucose moves

THANK YOU

Research prototype · not for clinical use

A twin that warns early, explains itself, and is honest about what it knows

Next: validate on real linked records from Indian clinics, with clinicians in the loop



github.com/AkhileshNalla11/glycaemic-digital-twin
Live demo and video links in the README